MIL-C-23280A(SHIPS)
26 November 1962
SUPERSEDING
MIL-C-23280(SHIPS)
11 May 1962

MILITARY SPECIFICATION

CABINET, ELECTRICAL EQUIPMENT, CY-2675

1. SCOPE

1.1 This specification covers the requirements for construction of an electrical equipment cabinet.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL
QQ-S-700 - Steel, Sheet and Strips, Medium Carbon.
QQ-S-763 - Steel Bars, Shapes, and Forgings, Corrosion Resisting.

MILITARY
MIL-E-16400 - Electronic Equipment, Naval Ship and Shore: General Specification.

STANDARDS
MIL-STD-189 - Racks, Electrical Equipment, 19-inch and Associated Panels.

DRAWINGS
RE10F775 - CY-2675 Cabinet, Electrical Equipment, Assembly and Details.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules

(Application for copies should be addressed to Official Classification Committee, 1 Park Avenue at 33d Street, New York 16, New York.)

3. REQUIREMENTS

3.1 Material.- Sheet metal shall be prime quality cold-rolled steel, dull finish, in accordance with QQ-S-700, unless otherwise noted on design drawing. Angles and shapes shall be prime quality, as specified in QQ-S-763. All screws, nuts, and washers shall be cadmium plated, in accordance with MIL-E-16400. All electrical connectors and hardware items shall be as specified in design drawing, or approved equals.

3.2 Construction.- Construction shall be in accordance with Drawing RE10F775.

3.3 Nomenclature.- Nomenclature of parts as used in this specification shall be as defined by Drawing RE10F775.

3.4 Finish.- Finish shall be in accordance with requirements of applicable paragraphs of MIL-E-16400 concerning preparation for, color of, and application of paint, except as otherwise noted on design drawing. Finished surface shall be free from scratches, paint runs, and other defects.

FSC 5975

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3.5 Welding.- All welded points shall conform to welding requirements of MIL-E-16400.

3.6 Breaking of corners.- There shall be no sharp corners or edges or any other condition which might be hazardous to personnel. Burrs and rough edges shall be ground smooth. Joints shall fit flush, with no cracks or gaps exceeding 1/16 inch in width.

3.7 Accommodation of standard panels.- The cabinet shall be capable of accommodating standard 19-inch panels and, therefore, dimensioning of tapped mounting holes on mounting angles shall conform to MIL-STD-189. The shape and dimensioning of the mounting angles whereon the standard panels are to be mounted shall be such that the panels may be easily installed.

3.7.1 Dimensioning of mounting holes and mounting angles.- For each mounting angle, the center of any mounting hole shall be within ±1/32 inch of a straight line between the centers of the top and bottom mounting holes of that mounting angle. The surfaces of each pair of mounting angles against which standard panels may be mounted and into which the mounting holes are tapped shall be; first, flat within +1/16 and -0 inch of a straight reference line; and second, parallel to a single plane surface within +1/16 and -0 inch.

3.7.2 Accuracy of location of mounting angles.- For each pair of mounting angles, diagonal distances between centers of uppermost mounting hole of each side and of lowermost mounting hole of opposite side shall differ by no more than 1/8 inch in order to assure squareness of assembly.

3.8 Squareness of overall assembly.- When the cabinet is sitting upright on a flat, horizontal surface, the diagonal distances from outer corner to outer corner of each face of the cabinet shall differ by no more than 1/8 inch to assure squareness of assembly.

3.9 Door assembly.- The front surface of the door shall deviate no more than 1/4 inch as measured in one direction from each of the two straight lines oriented diagonally from corner to corner of the door. Further, the front surface of the door shall not deviate more than 1/8 inch as measured in one direction from each of four straight lines extending along the front surface of the door 5/8 inch inward from the four edges. The surfaces around the outer edge of the door assembly shall not deviate more than 1/8 inch as measured in one direction from each of four straight lines along the outer surfaces at the sides, top, and bottom of the door.

3.10 Side and rear panel assemblies.- The panel assemblies shall meet the same requirements as does the door of 3.9, excluding the flanges at the top and bottom of the panel assemblies.

3.11 Assembly of cabinet.- Construction of the cabinet shall be such that it may be stored or shipped in a "knocked-down" condition, but easily assembled. When fully assembled, the cabinet shall be sturdy enough to withstand possible stresses and be rigid enough to preclude any difficulty of mounting standard racks due to deformation or any other reason.

3.12 Rigidity of overall assembly.- The cabinet, with no accessories in place, shall be supported so that it is balanced upon the bottom edge of one of its sides (not front or back). The diagonal distances from outer corner to outer corner of its front and back shall differ by no more than 1/8 inch from the measurements required in 3.8. The cabinet shall then be restored to an upright position and the diagonals shall again be measured; there shall be no evidence of permanent distortion.

3.13 Assembly instructions.- Each cabinet shall be furnished with instructions for assembly.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction inspection.- The preproduction model shall include the basic frame and any optional accessories listed in the contract or order. The preproduction inspection shall include applicable tests as

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specified in MIL-E-16400 and the following tests, except in the case where a particular optional accessory is not included in the procurement:

Subject	Paragraph
Surface examination	4.4.1
Assembly	4.4.2
Fit of standard panels	4.4.3.1
Spacing of tapped holes	4.4.3.2
Squareness of mounting assembly	4.4.3.3
Squareness of overall assembly	4.4.3.4
Electrical connectors	4.4.4
Rigidity of overall assembly	4.4.3.5
Side panel assembly	4.4.5.3
Rear panel assembly	4.4.5.4
Blower assembly	4.4.5.1
Door assembly	4.4.5.2
Bagged hardware	4.4.6

Any other inspection deemed necessary or included in the contract or order shall also be performed.

4.3 Sampling for quality conformance inspection. - Samples and lots for quality conformance inspection shall be as specified in the contract or order.

4.4 Quality conformance inspection. - Quality conformance inspection shall consist of the applicable tests of MIL-E-16400 as well as those listed in 4.2, unless otherwise specified (see 6.1). With the exception of the tests of 4.4.1 and 4.4.2, tests shall be performed upon the assembled cabinet.

4.4.1 Surface examination. - The cabinet shall be examined in accordance with MIL-E-16400 for workmanship, construction and fit, materials, parts, and finish, to assure compliance with 3.1, 3.3, 3.4, 3.5, and 3.6, insofar as possible by visual examination.

4.4.2 Assembly. - The unassembled cabinet shall be assembled as described in the assembly instructions. Any unreasonable difficulty in fitting of parts or installing of hardware, or any necessity for bending or otherwise reshaping any part, shall be cause for rejection, except that side or rear panel assemblies may be flexed to assure freedom from rattle.

4.4.3 Shapes and dimensions. - The assembled cabinet shall be subjected to the following tests:

4.4.3.1 Fit of standard panels. - A standard panel of size TC or UC, as defined in MIL-STD-189, shall be mounted, in turn, at the top, at the middle, and at the bottom of each pair of mounting angles by simultaneously inserting four screws or studs through the uppermost and lowermost holes of each side of the standard panel and into the corresponding tapped holes of the cabinet. If studs are used, their diameter shall be at least 0.10 inch. Alternately, studs of sufficient size to preclude their bending during this test may be permanently installed at the locations of the four holes in the standard panel, as described herein. At each location tested, alignment of the mounting holes shall be such that the screws or studs may be inserted with reasonable effort and the standard panel shall fit against the mounting angles in such a manner that no space between the standard panel and any mounting angle exceeds 1/16 inch.

4.4.3.2 Spacing of tapped holes. - Vertical and horizontal spacing of tapped holes on mounting angles shall be measured for tolerances as given by MIL-STD-189.

4.4.3.3 Squareness of mounting assembly. - Appropriate measurements shall be made to verify compliance with the requirements of 3.7.2.

4.4.3.4 Squareness of overall assembly. - Appropriate measurements shall be made to verify compliance with the requirements of 3.8.

4.4.3.5 Rigidity of overall assembly. - Appropriate measurements shall be to verify compliance with the requirements of 3.12.

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4.4.4 Electrical connectors.- Electrical power shall be applied to the cabinet and each outlet shall be tested by applying a load drawing at least 1 ampere. Excessive voltage drop or poor mechanical compatibility of outlets with plugs shall be cause for rejection.

4.4.5 Accessory assemblies.- The following tests shall be performed if the particular accessory assembly is included in the contract or order (see 6.1). Failure to pass any test shall be cause for rejection.

4.4.5.1 Blower assembly.- The entire duct assembly and its connection to the blower panel assembly should be examined for absence of gaps or cracks which might result in significant air leaks. With the flexible duct disconnected at any end, power shall be applied to the blower motor and operation of the blower verified.

4.4.5.2 Door assembly.- Measurements shall be taken to insure that the requirements of 3.9 have been met. Also, the door shall be assembled to the cabinet, and then opened and closed to verify proper operation of the hinge and latch. The fit of the door shall be such that it moves freely without any tendency to bind due to contact with other portions of the cabinet. When front and rear doors are specified in the contract or order, each door shall be tested for fit in both positions to assure interchangeability.

4.4.5.3 Side panel assembly.- Measurements shall be taken to insure compliance with the requirements of 3.10. Also, the panels shall be temporarily mounted to the cabinet to verify ease of installation and removal, proper fit, and firmness of position during conditions simulating normal shock and vibration.

4.4.5.4 Rear panel assembly.- Measurements shall be taken to insure compliance with the requirements of 3.10. Fit and firmness verification shall be the same as specified in 4.4.5.3.

4.4.6 Bagged hardware.- Verify that the proper hardware items are bagged and the bag attached to the base assembly, and if appropriate, that the proper hardware items are bagged and the bag attached to the door handle, both as detailed in drawing RE10F775.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation and for storage of onboard repair parts.-

5.1.1 Electrical equipment cabinet, CY-2675.

5.1.1.1 Preservation and packaging.- Preservation and packaging shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation and may conform to the suppliers commercial practice when such meets these requirements.

5.1.1.2 Packing.- Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the suppliers commercial practice when such meets these requirements.

5.1.1.3 Marking.- Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor s name and destination.

5.2 Domestic shipment and storage or overseas shipment.- The requirements and levels of preservation, packaging, packing, and marking for shipment shall be specified by the procuring activity (see 6.1).

(5.2.1 The following provides various levels for protection during domestic shipment and storage or overseas shipment which may be required when procurement is made (see 6.1):

5.2.1.1 Preservation and packaging, packing and marking.- The cabinet, basic frame and accessories, shall be preserved and packaged by level A or C, packed by level A, B, or C, and marked as specified (see 6.1) in accordance with MIL-E-17555.)

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6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Sizes of samples for quality conformance inspection (see 4.3).
(c) Added or deleted tests (see 4.4).
(d) Selection of subassemblies for quality conformance inspection (see 4.4.2).
(e) Quantity of each accessory (see 4.4.5).
(f) Preservation and packaging, packing, and marking requirements if other than as specified in 5.1 (see 5.2).

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in in any way be related thereto.

Preparing activity:
Navy - Ships
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BUDGET BU. NO. 45-R309

SPECIFICATION ANALYSIS SHEET
NAVSHIPS-4863 (8-61)

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Bureau of Ships

This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this form will be appreciated.

Fold on dotted lines on reverse side, staple in corner, and send to Bureau of Ships, Specifications and Standardization Branch, Washington 25, D.C.

SPECIFICATION

ORGANIZATION	CITY	STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A DIRECT GOVERNMENT CONTRACT ☐ OR A SUBCONTRACT ☐

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   a. GIVE PARAGRAPH NUMBER AND WORDING

   b. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID.

3. IS THE SPECIFICATION RESTRICTIVE? ☐ YES ☐ NO IF THE ANSWER IS "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification.)* PLACE THIS FORM AND PAPERS IN AN ENVELOPE AND SEND TO THE BUREAU.

SUBMITTED BY *(Print name and activity)*	DATE

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